Experiment #5
Test of Hypothesis and Confidence intervals – Z-tests and t-test

Course Outcome

Aim:
To understand the testing of hypothesis for large and small sample using R functions

Testing of Hypothesis - Large Sample Mean Test

Hypothesis tests are used to make decisions or judgments about the value of a parameter, such as the population mean. There are two approaches for conducting a hypothesis test: the critical value approach and the P-value approach. Since a sample statistic is used, it is possible that the decision reached is an error. There are two types of errors: Type I Error and Type II Error.

Test of significance of the difference between sample mean and population mean

$$z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

Procedure

1. Import the data set.
2. Determine the critical value and sample statistic using R functions.
3. Conclude the problem using R functions.

Problem
Suppose the mean weight of King Penguins found in an Antarctic colony last year was 15.4 kg. In a sample of 35 penguins same time this year in the same colony, the mean penguin weight is 14.6 kg. Assume the population standard deviation is 2.5 kg. At 0 .05 significance level, can we reject the null hypothesis that the mean penguin weight does not differ from last year?

Code and Results:

```
# Input the sample mean
 xbar=14.6
 xbar

## [1] 14.6
```

```
# Input the population mean
mu0=15.4
mu0

## [1] 15.4

 # Input the standard deviation
sigma=2.5
sigma

## [1] 2.5

# Input the sample size
n=35
n

## [1] 35

# Test Statistic
 z=(xbar-mu0)/(sigma/sqrt(n))
 z

## [1] -1.893146

# Level of significance
alpha=0.05
alpha

## [1] 0.05

# Two-tailed critical value
zhalfalpha=qnorm(1-(alpha/2))
 zhalfalpha

 ## [1] 1.959964

c(-zhalfalpha,zhalfalpha)

 ## [1] -1.959964 1.959964

# To find p-value pval=2*pnorm(z) pval
pval = 2 * (1 - pnorm(abs(z)))

## [1] 0.05833852

# conclusion if(pval>alpha){print("Accept Null hypothesis")} else{print("Reject Null
hypothesis")}
```

Test of significance of the difference between sample proportion and population proportion

$$z = \frac{p - P}{\sqrt{\frac{PQ}{n}}}$$

where p is the sample proportion, P is the population proportion, $Q = 1 - P$, and n is the sample size.

Procedure:

- Import the data set.
- Determine the critical value and sample statistic using R functions.
- Conclude the problem using R functions.

Problem:

The fatality rate of typhoid patients is believed to be 17.26%. In a certain year 640 patients suffering from typhoid were treated in a metropolitan hospital and only 63 patients died. Can you consider the hospital efficient?

Code and Results:

```
# Input the data
# Size of the sample
 n=640
 n

## [1] 640

# Sample proportion
Sprop=63/n
Sprop

## [1] 0.0984375

# Population proportion
Pprop=0.1726
Pprop

## [1] 0.1726

# probability of failure
q=1-Pprop
q

## [1] 0.8274

# test statistic
z=(Sprop-Pprop)/sqrt(Pprop*q/n)
z
```

```
## [1] -4.964736

#critical value
 E=qnorm(.975)
 E

# critical region
c(-E,E)

## [1] -1.959964 1.959964

# confidence interval
Sprop+c(-E,E)*sqrt(Pprop*(1-Pprop)/n)

## [1] 0.06915985 0.12771515

# Conclusion
if(z>-E && z<E) {print("Hospital is not efficient")} else{print("Hospital is
efficient")}

## [1] "Hospital is efficient"
```

Test of significance of the difference between two sample means

$$z = \frac{\overline{X_1} - \overline{X_2}}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

Procedure:

- Import the data set.
- Determine the critical value and sample statistic using R functions.
- Conclude the problem using R functions

Problem:

In a random sample of size 500, the mean is found to be 20. In another independent sample of size 400, the mean is 15. Could the samples have been drawn from the same population with S.D 4?

Code and Results:

```
# Input the sample mean
 xbar=20
 xbar

## [1] 20

ybar=15
```

```
ybar

## [1] 15

# Input the standard deviation
sigma=4
sigma

## [1] 4

# Input the sample size
n1=500
n1

## [1] 500

n2=400
n2

## [1] 400

# Test Statistic
z=(xbar-ybar)/(sigma*sqrt((1/n1)+(1/n2)))
z

## [1] 18.6339

# Level of significance
alpha=0.05
alpha

## [1] 0.05

# Two-tailed critical value
zalpha=qnorm(1-(alpha/2))
zalpha

## [1] 1.959964

# conclusion if(z<=zalpha){print("Accept Null hypothesis")} else{print("Reject Null
hypothesis")}

## [1] "Reject Null hypothesis"
```

Test of significance of the difference between two sample proportions

$$z = \frac{p_1 - p_2}{\sqrt{PQ\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

Procedure:

- Import the data set.
- Determine the critical value and sample statistic using R functions.
- Conclude the problem using R functions

Problem:

In a large city A, 20% of a random sample of 900 school boys had a slight physical defect. In another large city B, 18.5% of a random sample of 1600 school boys had the same defect. Is the difference between the proportions significant?

Code and Results:

```
# Input the sample proportions
p1=0.20
p1

## [1] 0.2

p2=0.185
p2

## [1] 0.185

# Input the sample size
n1=900
n1

## [1] 900

n2=1600
n2

## [1] 1600

# To find approximate population proportion
P=(n1*p1+n2*p2)/(n1+n2)
P

## [1] 0.1904
```

```
Q=1-P
# Test Statistic
z=(p1-p2)/sqrt(P*Q*sqrt((1/n1)+(1/n2)))
z

## [1] 0.1871665

# Level of significance
alpha=0.05
alpha

## [1] 0.05

# Two-tailed critical value
zalpha=qnorm(1-(alpha/2))
zalpha

## [1] 1.959964

# conclusion
if(z<=zalpha){print("Accept Null hypothesis")} else{print("Reject Null hypothesis")}

## [1] "Accept Null hypothesis"
```

Testing of Hypothesis - Small Sample Mean Test

If the sample size is less than 30 i.e., n < 30, the sample may be regarded as small sample.

Student's t-test:

The student's t-test is mentioned as a method of testing the theory about the mean of a small sample drawn from a normally distributed population where the standard deviation of the given population is unknown. The t distribution belonging under a family of curves in which the number of degrees of freedom specifies a particular curve. As the sample size (and the degrees of freedom) increases, the t distribution approaches the bell shape of the standard normal distribution. In common, for tests involving the mean of a sample of size greater than 30, then the normal distribution is applied.

$$t = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

Paired t-test:

The paired t-test is a method used to test whether the mean difference between pairs of measurements is zero or not. We can use the test when your data values are paired measurements. For example, you might have before-and-after measurements for a group of people. Also, the distribution of differences between the paired measurements should be normally distributed.

$$t = \frac{\mu_d}{s/\sqrt{n}}$$

Problem: (Student's t-test)

Two independent samples of sizes 8 and 7 contained the following values:

Sample 1	19	17	15	21	16	18	16	14
Sample 2	15	14	15	19	15	18	16	20

Is the difference between the sample means significant?

Code and Results:

```
# Problem 1
# input the data
sample1=c(19,17,15,21,16,18,16,14)
sample1

## [1] 19 17 15 21 16 18 16 14

sample2=c(15,14,15,19,15,18,16,20)
sample2

## [1] 15 14 15 19 15 18 16 20

# output using t-distribution
t=t.test(sample1, sample2, var.=TRUE)
t

          #Two Sample t-test

#data:  sample1 and sample2
#t = 0.44721, df = 14, p-value =
#0.6616
#alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
# -1.897944  2.897944
#sample estimates:
#mean of x mean of y
    17.0      16.5
```

```
# test-statistic
cv=t$statistic
cv

## t
## 0.4472136

#critical value
tv=qt(0.975,14)
tv

## [1] 2.144787

#conclusion
if(cv <= tv){print("Accept Ho")} else{print("Reject Ho")}

## [1] "Accept Ho"
```

Problem: (Paired t-test)

The following data relate to the marks obtained by 10 students in two tests; one held at the beginning of a year and the other at the end of the year after intensive coaching. Do the data indicate that the students have got benefited by coaching?

Test 1	19	17	15	21	16	18	16	14	19	20
Test 2	15	14	15	19	15	18	16	20	22	19

```
#Paired- t-test
# input the data
test1=c(19,17,15,21,16,18,16,14,19,20)
test1

## [1] 19 17 15 21 16 18 16 14 19 20

test2=c(15,14,15,19,15,18,16,20,22,19)
test2

## [1] 15 14 15 19 15 18 16 20 22 19

t=t.test(sample1,sample2,paired=TRUE)
t
```

```
##
## Paired t-test
##
## data: sample1 and sample2
## t = 0.46771, df = 7, p-value = 0.6542
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
## -2.02789 3.02789
## sample estimates:
## mean of the differences
## 0.5

# level of significance
alpha=0.05
# p-value
tv=t$p.value
tv
## [1] 0.6542055
# conclusion
if(tv >alpha){print("Accept Ho")} else{print("Reject Ho")}

## [1] "Accept Ho
```

Conclusion:

Testing of hypothesis for large and small sample tests using R functions has been explored and concluded.